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DEEP NEURAL NETWORK MODULE # 8: ATTENTION MECHANISM

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The instructor is gratefully acknowledging
the authors who made their course
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This deck is prepared by Seetha Parameswaran.

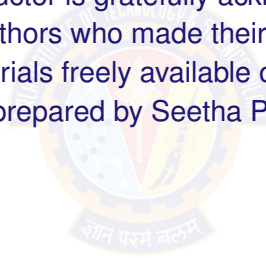


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WHY STUDY ENCODER-DECODER?

Many NLP tasks require variable-length sequence-to-sequence mapping:

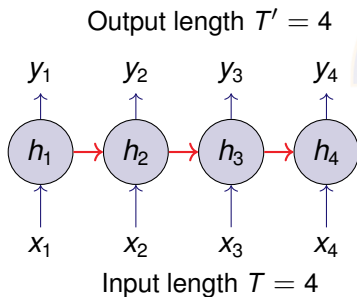
- Sentiment Analysis:
 - ▶ Input: "The phone battery lasts only 2 hours. Very disappointed."
 - ▶ Output: "Negative"
- Question Answering:
 - ▶ Input: Context + "What is the capital of India?"
 - ▶ Output: "New Delhi"
- Text Summarization:
 - ▶ Input: 500-word article
 - ▶ Output: 50-word summary

Key observation: Input length $T \neq$ Output length T' in general.

THE CHALLENGE: STANDARD RNN LIMITATIONS

Example: Question Answering fails with standard RNN

Standard RNN: Produces one output per input step (requires $T = T'$)



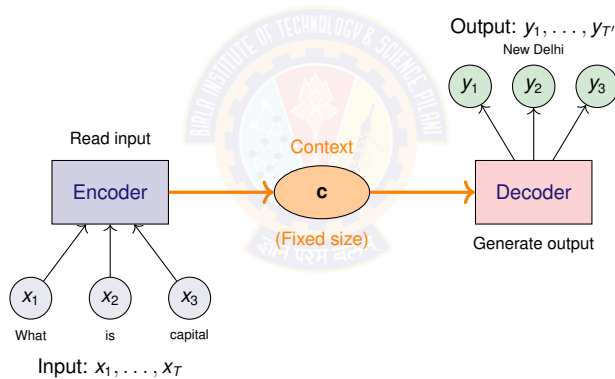
- Input: "India gained independence from Britain. What year was independence?" (10 words)
- Desired Output: "1947" (1 word)
- Problem: Standard RNN produces 10 outputs, not 1!

Solution needed: Separate **reading** phase from **generation** phase.

SOLUTION: ENCODER-DECODER ARCHITECTURE

Key Idea: Two-stage process

- 1 **Encoder:** Read and compress entire input into context vector $\mathbf{c}$
- 2 **Decoder:** Generate output sequence from context, one token at a time



Flexibility: T (input length) can differ from T' (output length).

HOW ENCODER-DECODER WORKS

- Phase 1: Encoding

- ▶ Process input sequence with RNN:

$$h_t^{\text{enc}} = \phi(W_{hh}^{\text{enc}} h_{t-1}^{\text{enc}} + W_{hx}^{\text{enc}} x_t + b_h^{\text{enc}})$$

- ▶ Extract context from final hidden state:

$$\mathbf{c} = h_T^{\text{enc}}$$

- Phase 2: Decoding

- ▶ Initialize decoder with context:

$$h_0^{\text{dec}} = \mathbf{c}$$

- ▶ Generate output sequence autoregressively:

$$h_t^{\text{dec}} = \phi(W_{hh}^{\text{dec}} h_{t-1}^{\text{dec}} + W_{hy}^{\text{dec}} y_{t-1} + b_h^{\text{dec}})$$

- ▶ Output probability distribution:

$$P(y_t | y_{<t}, \mathbf{c}) = \text{softmax}(W_o h_t^{\text{dec}} + b_o)$$

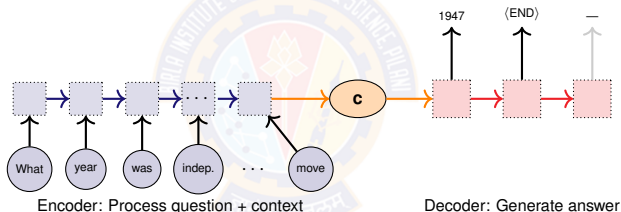
EXAMPLE: QUESTION ANSWERING WITH ENCODER-DECODE

- **Task:** Answer question from context

Context: "India gained independence in 1947. Gandhi led the movement."

Question: "What year was independence?"

Answer: "1947"

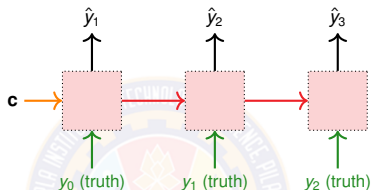


- **Process:**

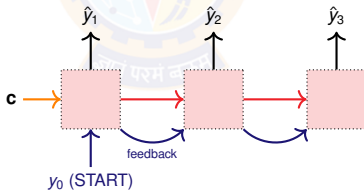
- ▶ Encoder reads entire input ($T \approx 15$ tokens)
- ▶ Context **c** encodes: "answer is about year and independence"
- ▶ Decoder generates: "1947" then <END> ($T' = 1$ meaningful token)

TRAINING VS INFERENCE

- Training (Teacher Forcing): Use ground truth as decoder input



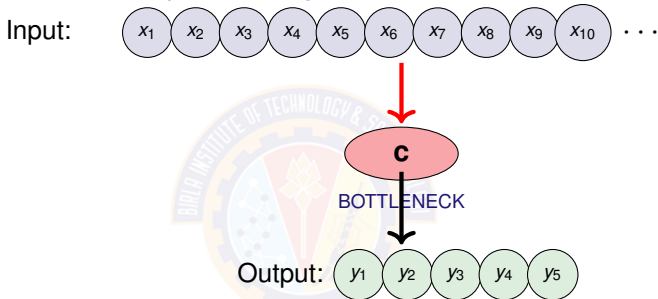
- Inference (Free Running): Use model's own predictions



- Exposure Bias: Decoder trained on ground truth but tested on own predictions!

THE CRITICAL PROBLEM: CONTEXT BOTTLENECK

- All input information must pass through fixed-size vector $\mathbf{c}$.



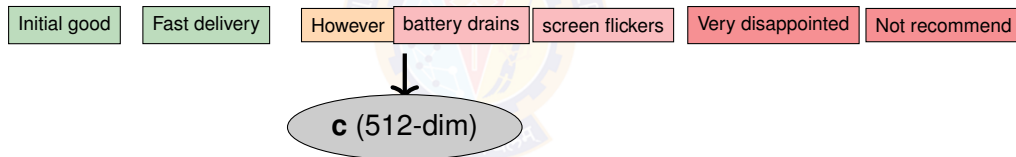
- Problems:
 - ① **Information loss:** Long sequences $\rightarrow$ early tokens forgotten
 - ② **Fixed capacity:** Vector size (e.g., 512-dim) limits information storage
 - ③ **Uniform encoding:** All input parts weighted equally, no selectivity
- **Example:** 50-word review compressed to 512 numbers, then generate sentiment!

BOTTLENECK IMPACT: CONCRETE EXAMPLE

- **Task:** Sentiment analysis of product review

- **Input (30 words):**

"Ordered this laptop for work. Initial impressions were good. Fast delivery. However, after 2 weeks, battery drains in 2 hours. Screen flickers constantly. Very disappointed. Would not recommend."

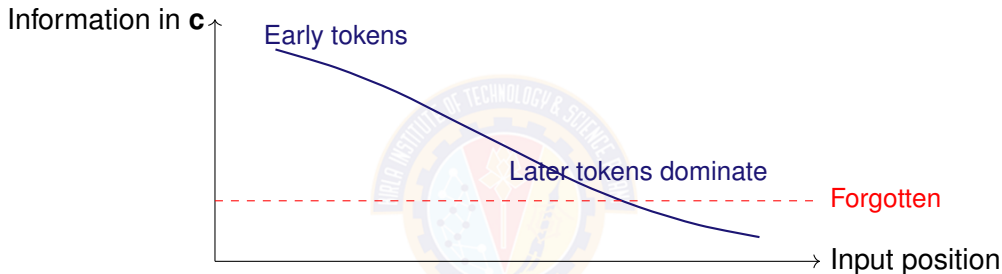


Early positive sentiment lost?
Key phrases "very disappointed" weakened?

- **Issue:** When generating "Negative", decoder needs to emphasize later parts, but **c** treats all parts equally!

WHY FIXED CONTEXT FAILS FOR LONG SEQUENCES

Visualization of information loss:



Observations from research (Sutskever et al., 2014):

- Performance degrades significantly for sequences > 30 tokens
- First few words often lost by the time decoder generates output
- Model struggles with long-range dependencies

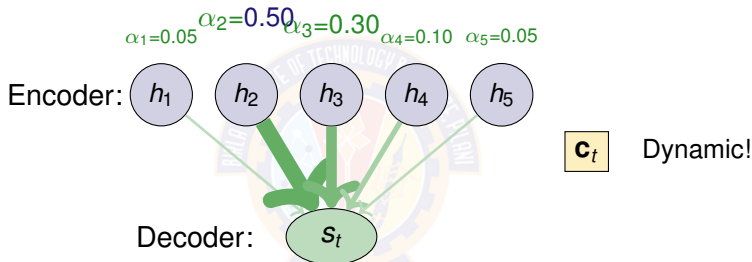
Question: How can we avoid this bottleneck?

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THE SOLUTION: ATTENTION MECHANISMS

- **Core Insight:** Instead of single fixed context, let decoder **selectively attend** to relevant input parts at each step.



- **Key Differences from Encoder-Decoder:**
 - ▶ **Before:** Single context $\mathbf{c}$ for entire decoding
 - ▶ **After:** Different context $\mathbf{c}_t$ at each decoder step t
 - ▶ Weights α_j learned automatically from data
 - ▶ Focuses on relevant input positions dynamically

HISTORICAL CONTEXT: THE BREAKTHROUGH

Timeline:

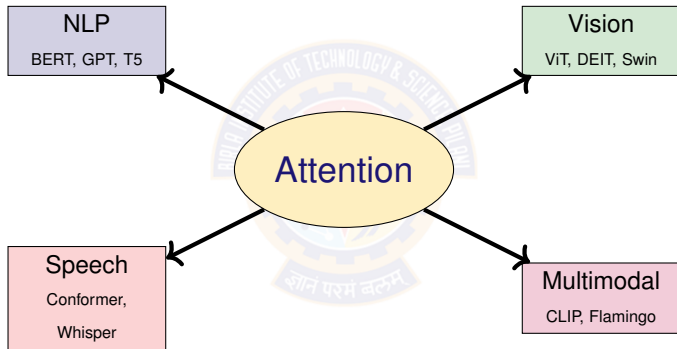


Impact of Bahdanau et al. (2014):

- Introduced attention for neural machine translation
- Improved BLEU scores significantly on long sentences
- Decoder could "align" to relevant source words
- Laid foundation for Transformer architecture (2017)

MODERN LANDSCAPE: ATTENTION EVERYWHERE

Attention has become the dominant mechanism across AI domains.



Understanding attention is essential for modern deep learning.

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ATTENTION

DEFINITION

Attention is a mechanism that computes a weighted sum of values based on the similarity between a query and keys.

Components:

- **Query ($\mathbf{q}$):** What we're looking for (decoder state)
- **Keys ($\{\mathbf{k}_i\}$):** What's available to match (encoder states)
- **Values ($\{\mathbf{v}_i\}$):** Actual content to retrieve (encoder states)

Mathematical Form:

$$\text{Attention}(\mathbf{q}, \{\mathbf{k}_i, \mathbf{v}_i\}_{i=1}^n) = \sum_{i=1}^n \alpha_i \mathbf{v}_i$$

where attention weights:

$$\alpha_i = \frac{\exp(\text{score}(\mathbf{q}, \mathbf{k}_i))}{\sum_{j=1}^n \exp(\text{score}(\mathbf{q}, \mathbf{k}_j))} \quad (\text{softmax normalization})$$

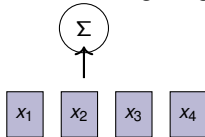
CORE CONCEPT: WEIGHTED COMBINATIONS

Key Principle: Not all inputs contribute equally – attention learns relevance.

Without Attention:

$$\text{output} = f \left(\sum_{i=1}^n x_i \right)$$

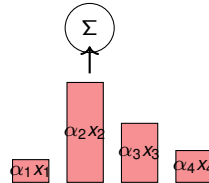
Uniform weighting



With Attention:

$$\text{output} = f \left(\sum_{i=1}^n \alpha_i x_i \right)$$

Learned weights α_i

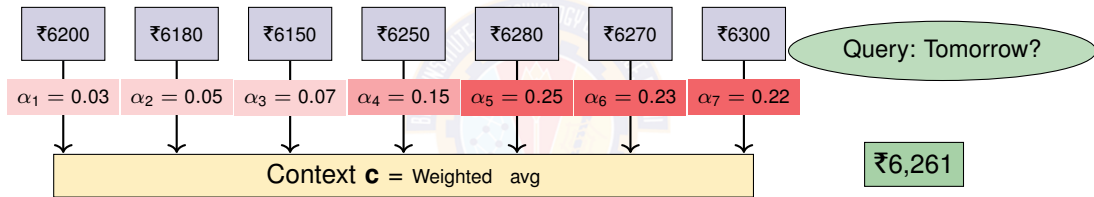


Constraint: Weights sum to 1: $\sum_{i=1}^n \alpha_i = 1$ (probability distribution)

EXAMPLE 1: STOCK PRICE PREDICTION

Task: Predict tomorrow's gold price (₹/gram) using past 7 days.

Input: Gold prices = [₹6,200, ₹6,180, ₹6,150, ₹6,250, ₹6,280, ₹6,270, ₹6,300]



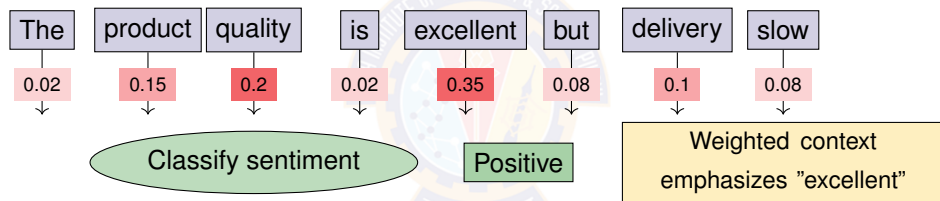
Observation: Recent days (5-7) receive higher attention weights α_i .

$$\begin{aligned}\mathbf{c} &\approx 0.03 \times 6200 + 0.05 \times 6180 + 0.07 \times 6150 + 0.15 \times 6250 + \\ &\quad 0.25 \times 6280 + 0.23 \times 6270 + 0.22 \times 6300 \\ &\approx 6261\end{aligned}$$

EXAMPLE 2: SENTIMENT ANALYSIS

Task: Classify review sentiment while identifying key phrases.

Input: "The product quality is excellent but delivery was very slow"

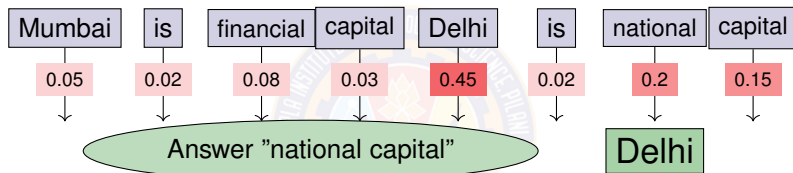


Key Insight: Model learns "excellent" ($\alpha_5 = 0.35$) is most important for sentiment, despite negative word "slow" present.

EXAMPLE 3: Q& A WITH ATTENTION

Context: "Mumbai is the financial capital. Delhi is the national capital."

Question: "What is the national capital?"

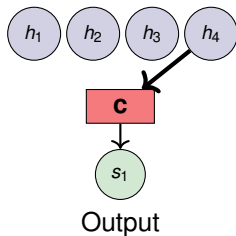


Attention weights show:

- Highest attention on "Delhi" ($\alpha_5 = 0.45$)
- Secondary attention on "national" ($\alpha_7 = 0.20$) and "capital" ($\alpha_4 = 0.15$)
- Model correctly ignores "financial capital"

WITHOUT VS WITH ATTENTION: SIDE-BY-SIDE

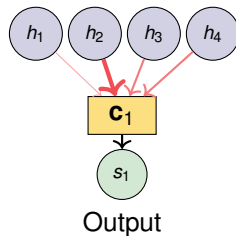
Without Attention:



Issues:

- Fixed bottleneck c
- Information loss
- Poor long sequences

With Attention:



Benefits:

- Dynamic context c_t
- Selective focus
- Handles long sequences

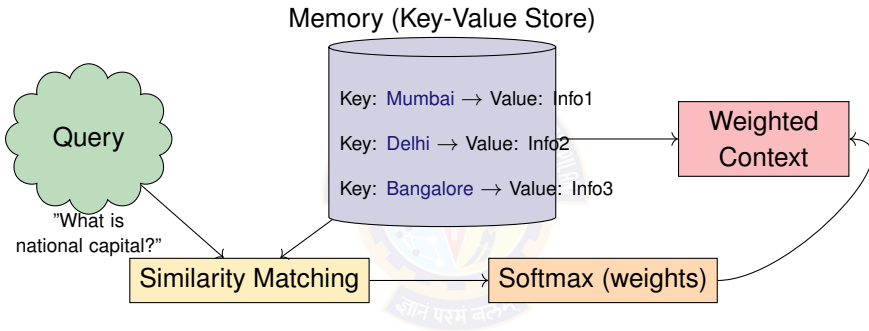
Key Difference: Context adapts to what decoder needs at each step!

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ATTENTION AS INFORMATION RETRIEVAL

Analogy: Attention works like a database lookup system.



Process:

- 1 Query searches memory via **key similarity**
- 2 Compute attention weights based on query-key match
- 3 Retrieve **values** weighted by attention

QUERY-KEY-VALUE (QKV): COMPONENTS

Three distinct roles in attention mechanism:

- Query ($\mathbf{q}$):
 - ▶ What information are we seeking?
 - ▶ Comes from decoder state: "I need to generate next word"
 - ▶ Example: Decoder state when generating "Delhi"
- Keys ($\{\mathbf{k}_i\}$):
 - ▶ What's available to match against?
 - ▶ Comes from encoder states: representations of input tokens
 - ▶ Example: Encoded representations of [Mumbai, is, financial, capital, Delhi, ...]
- Values ($\{\mathbf{v}_i\}$):
 - ▶ What actual content do we retrieve?
 - ▶ Also from encoder states (often same as keys)
 - ▶ Example: Contextual embeddings of input words

Intuition: Query searches Keys, retrieves matching Values.

QKV PARADIGM: DETAILED MECHANICS

Step-by-step attention computation:

- 1 Compute compatibility scores between query and all keys:

$$e_i = \text{score}(\mathbf{q}, \mathbf{k}_i) \quad \text{for } i = 1, \dots, n$$

- 2 Normalize with softmax to get attention weights:

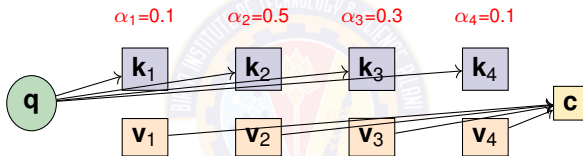
$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^n \exp(e_j)} \quad (\text{ensures } \sum_i \alpha_i = 1)$$

- 3 Compute weighted sum of values:

$$\mathbf{c} = \sum_{i=1}^n \alpha_i \mathbf{v}_i$$

QKV PARADIGM: VISUAL

- Query $\mathbf{q}$: What information are we seeking? Decoder state
- Key $\mathbf{k}_i$: What's available to match against? Encoder state
- Value $\mathbf{v}_i$: What actual content do we retrieve? Encoder state

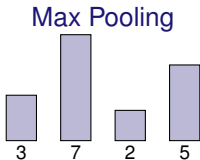


$$\alpha_i = \text{softmax}(\text{score}(\mathbf{q}, \mathbf{k}_i)) \quad \text{for } i = 1, \dots, n$$

$$\mathbf{c} = \sum_{i=1}^n \alpha_i \mathbf{v}_i$$

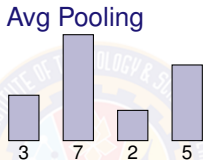
ATTENTION AS WEIGHTED POOLING

Pooling Perspective: Attention is soft, adaptive pooling over sequence.



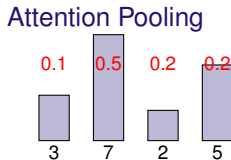
max = 7

Hard selection



avg = 4.25

Uniform weights



$$\sum \alpha_i v_i$$

Learned weights

Key Difference:

- Max/Avg pooling: Fixed, uniform aggregation
- Attention: **Data-dependent, learned** aggregation based on query

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DOT-PRODUCT ATTENTION

Simplest form: Measure similarity via inner product.

$$\text{score}(\mathbf{q}, \mathbf{k}_i) = \mathbf{q}^\top \mathbf{k}_i = \sum_{j=1}^d q_j k_{i,j}$$

$$\text{Attention } \alpha_i = \frac{\exp(\mathbf{q}^\top \mathbf{k}_i)}{\sum_{j=1}^n \exp(\mathbf{q}^\top \mathbf{k}_j)} = \text{softmax}(\mathbf{q}^\top \mathbf{K}^\top)_i$$

where $\mathbf{K} = [\mathbf{k}_1, \dots, \mathbf{k}_n]^\top \in \mathbb{R}^{n \times d}$.

EXAMPLE: DOT-PRODUCT ATTENTION

Example: Stock price prediction

- Query: $\mathbf{q} = [0.8, 0.6]^\top$ (tomorrow's expected trend features)
- Keys: $\mathbf{k}_1 = [0.5, 0.3]^\top$, $\mathbf{k}_2 = [0.9, 0.7]^\top$, $\mathbf{k}_3 = [0.4, 0.2]^\top$
- Scores:

$$e_1 = 0.8(0.5) + 0.6(0.3) = 0.58$$

$$e_2 = 0.8(0.9) + 0.6(0.7) = 1.14 \quad (\text{highest!})$$

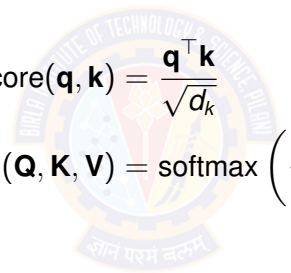
$$e_3 = 0.8(0.4) + 0.6(0.2) = 0.44$$

- Weights: $\alpha_1 = 0.21$, $\alpha_2 = 0.62$, $\alpha_3 = 0.17$ (after softmax)
- Interpretation: Day 2 most aligned with query $\rightarrow$ highest attention.

SCALED DOT-PRODUCT ATTENTION

Problem: For large dimension d , dot products grow large $\rightarrow$ softmax saturates.

Solution (Vaswani et al., 2017): Scale by $\sqrt{d_k}$ to normalize variance.


$$\text{score}(\mathbf{q}, \mathbf{k}) = \frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$$

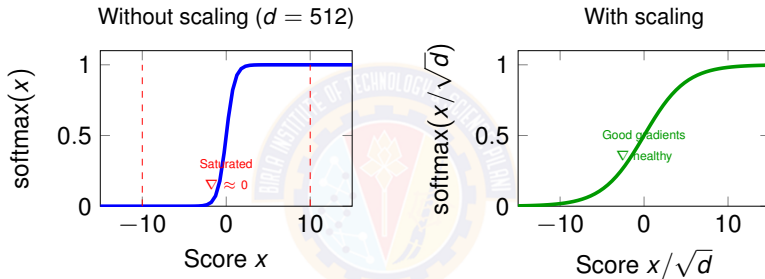
$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}} \right) \mathbf{V}$$

Benefits:

- Keeps scores in reasonable range regardless of dimension
- Prevents vanishing gradients in softmax
- Standard in all Transformer models

WHY SCALING MATTERS: GRADIENT FLOW

Effect of scaling on softmax gradients:



Impact:

- **Left:** Large scores push softmax to extremes (0 or 1)
- **Right:** Scaled scores maintain gradient flow
- Critical for training deep networks (6-24+ layers)

ADDITIVE ATTENTION (BAHDANAU, 2014)

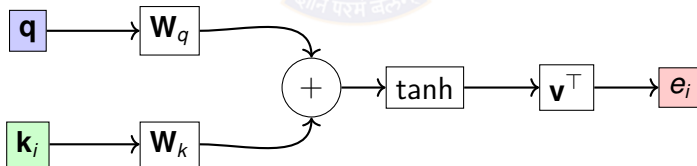
Original attention mechanism for neural machine translation.

$$\text{score}(\mathbf{q}, \mathbf{k}_i) = \mathbf{v}^\top \tanh(\mathbf{W}_q \mathbf{q} + \mathbf{W}_k \mathbf{k}_i)$$

where:

- $\mathbf{W}_q \in \mathbb{R}^{h \times d_q}$: Query projection matrix
- $\mathbf{W}_k \in \mathbb{R}^{h \times d_k}$: Key projection matrix
- $\mathbf{v} \in \mathbb{R}^h$: Learnable weight vector
- h : Hidden dimension (hyperparameter, e.g., 256)

Computation flow:



COMPARISON: DOT-PRODUCT VS ADDITIVE

Aspect	Dot-Product (Scaled)	Additive (Bahdanau)
Formula	$\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$	$\mathbf{v}^\top \tanh(\mathbf{W}_q \mathbf{q} + \mathbf{W}_k \mathbf{k})$
Parameters	None	$\mathbf{W}_q, \mathbf{W}_k, \mathbf{v}$
Complexity	$O(d)$ per pair	$O(d \cdot h + h)$ per pair
Expressiveness	Limited to linear	Nonlinear (tanh)
Parallelization	Excellent (matrix mult)	Moderate
Dimension req.	Requires $d_q = d_k$	Works with $d_q \neq d_k$
Modern usage	Transformers	Early seq2seq

Historical note: Additive was better for RNN-based models (2014-2016), but scaled dot-product dominates in Transformer era (2017+).

COMPUTATIONAL COMPLEXITY ANALYSIS

For sequence length n , dimension d , and m queries:

Operation	Scaled Dot-Product	Additive
Score computation	$O(mnd)$	$O(mndh + mnh)$
Softmax	$O(mn)$	$O(mn)$
Weighted sum	$O(mnd)$	$O(mnd)$
Total	$O(mnd)$	$O(mndh)$

Matrix formulation (batch processing):

- Scaled dot-product: $\mathbf{QK}^T \in \mathbb{R}^{m \times n}$ computed in one GPU operation
- Highly parallelizable on modern hardware (GPUs/TPUs)
- Memory efficient: $O(mn)$ for attention matrix

Practical impact: Dot-product is 3-5× faster in practice due to better hardware utilization.

SCORING FUNCTIONS: OVERVIEW

The scoring function $\text{score}(\mathbf{q}, \mathbf{k})$ measures query-key compatibility.

Method	Score Function	Parameters
Dot Product	$\mathbf{q}^\top \mathbf{k}$	None
Scaled Dot Product	$\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}$	None
Additive (Bahdanau)	$\mathbf{v}^\top \tanh(\mathbf{W}_q \mathbf{q} + \mathbf{W}_k \mathbf{k})$	$\mathbf{W}_q, \mathbf{W}_k, \mathbf{v}$
Multiplicative (Luong)	$\mathbf{q}^\top \mathbf{W} \mathbf{k}$	$\mathbf{W}$

Design Trade-offs:

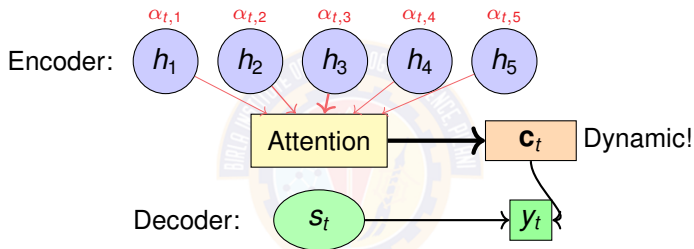
- **Dot product:** Fast (matrix ops), no parameters, requires $d_q = d_k$
- **Scaled Dot Product :** Prevents saturation for large dimensions (Transformer default)
- **Additive:** More expressive, flexible dimensions, more parameters

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BAHDANAU ATTENTION: THE BREAKTHROUGH

Key Innovation (Bahdanau et al., 2014): Replace fixed context with dynamic attention.



Context at time t :

$$\mathbf{c}_t = \sum_{i=1}^T \alpha_{t,i} \mathbf{h}_i$$

where $\alpha_{t,i}$ depends on current decoder state s_{t-1} and encoder state h_i .

BAHDANAU ATTENTION: MATH

At each decoder time step t :

- Step 1: Compute alignment scores
- Step 2: Normalize to attention weights
- Step 3: Compute context vector
- Step 4: Update decoder state
- Step 5: Generate output distribution

$$e_{t,i} = \mathbf{v}^\top \tanh(\mathbf{W}_s \mathbf{s}_{t-1} + \mathbf{W}_h \mathbf{h}_i)$$

$$\alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_{j=1}^T \exp(e_{t,j})}$$

$$\mathbf{c}_t = \sum_{i=1}^T \alpha_{t,i} \mathbf{h}_i$$

$$\mathbf{s}_t = \text{RNN}([y_{t-1}; \mathbf{c}_t], \mathbf{s}_{t-1})$$

$$P(y_t | y_{<t}, \mathbf{x}) = \text{softmax}(\mathbf{W}_o [\mathbf{s}_t; \mathbf{c}_t])$$



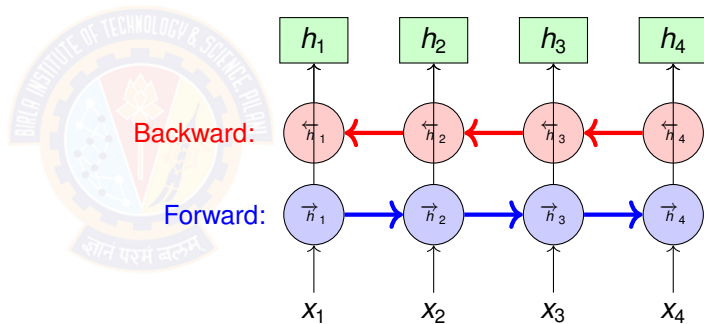
ENCODER WITH ATTENTION: BIDIRECTIONAL RNN

Bahdanau used bidirectional RNN encoder to capture context from both directions.

$$\vec{h}_i = \text{RNN}_{\text{fwd}}(x_i, \vec{h}_{i-1})$$

$$\overleftarrow{h}_i = \text{RNN}_{\text{bwd}}(x_i, \overleftarrow{h}_{i+1})$$

$$h_i = [\vec{h}_i; \overleftarrow{h}_i]$$



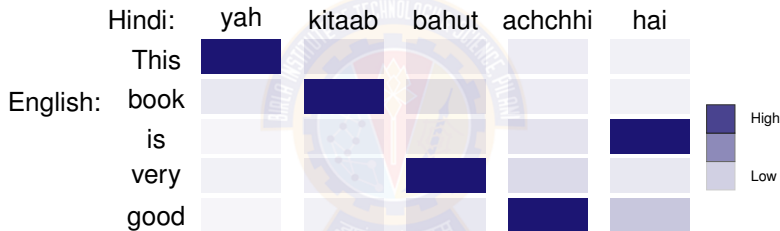
Benefit: h_i encodes both past and future context for position i .

ATTENTION ALIGNMENT: VISUALIZATION

Example: Hindi to English translation

Source (Hindi): yah kitaab bahut achchhi hai

Target (English): This book is very good



Observations:

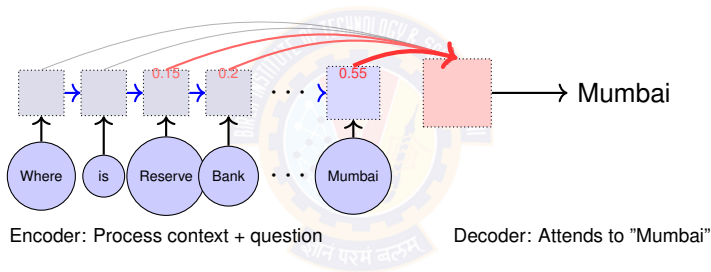
- Strong diagonal pattern: word-level alignment
- "yah" $\leftrightarrow$ "This": $\alpha = 0.85$
- "kitaab" $\leftrightarrow$ "book": $\alpha = 0.82$

EXAMPLE: Q& A WITH ATTENTION

Context: "Mumbai is the financial capital. The Reserve Bank is located there."

Question: "Where is Reserve Bank?"

Answer: "Mumbai"



Attention interpretation:

- Decoder focuses on context mentioning "Reserve Bank" and "Mumbai"
- Highest weight ($\alpha = 0.55$) on "Mumbai" in context
- Successfully extracts answer from relevant part of input

IMPLEMENTATION CONSIDERATIONS

- Training with Attention:

- ▶ Teacher forcing: Use ground-truth y_{t-1} as decoder input
- ▶ Loss function: Cross-entropy over target vocabulary

$$\mathcal{L} = - \sum_{t=1}^{T'} \log P(y_t^* | y_{<t}^*, \mathbf{x})$$

- ▶ Optimizer: Adam with learning rate warmup and decay
- ▶ Regularization: Dropout on encoder/decoder states and attention weights

- Inference with Attention:

- ▶ Greedy decoding: Select $\arg \max_y P(y_t | y_{<t}, \mathbf{x})$ at each step
- ▶ Beam search: Maintain top- k hypotheses (typical $k = 5$)
- ▶ Attention recomputed: Fresh context $\mathbf{c}_t$ at each decoder step
- ▶ Stopping: Generate until $\langle \text{END} \rangle$ token or max length

- Computational cost:

- ▶ Attention: $O(T \times T')$ for input length T , output length T'

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ATTENTION TYPES: TAXONOMY

Type	Q Source	K,V Source	Use Case
Self-Attention	Same sequence	Same sequence	Intra-sequence relationships
Cross-Attention	Sequence A	Sequence B	Inter-sequence alignment (seq2seq)
Masked Attention	Same sequence	Same (causal mask)	Autoregressive generation
Multi-Head	Same (parallel)	Same (parallel)	Multiple relationship types

Key Distinctions:

- **Source of Q, K, V:** Where do query, key, value vectors come from?
- **Masking:** What positions can attend to what?
- **Purpose:** What relationships are being modeled?

SELF-ATTENTION: DEFINITION AND PURPOSE

- **Self-attention:** Query, key, and value all come from the *same* sequence.

$$\mathbf{Q} = \mathbf{XW}_Q, \quad \mathbf{K} = \mathbf{XW}_K, \quad \mathbf{V} = \mathbf{XW}_V$$

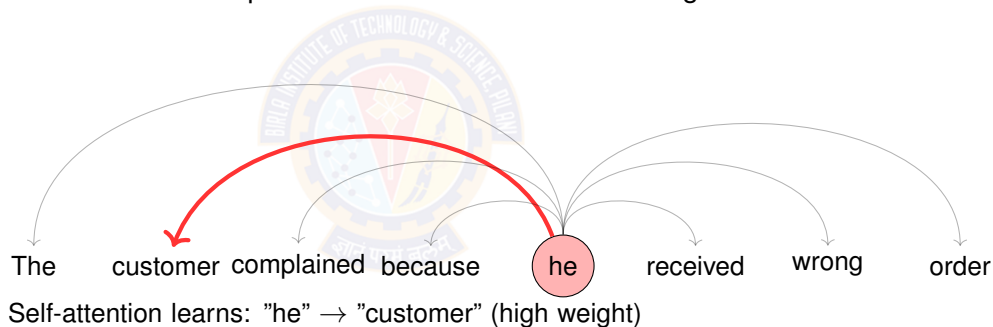
where $\mathbf{X} = [x_1, \dots, x_n]^\top$ is the input sequence.

- **Purpose:** Model relationships *within* a sequence.
- **Time Series Example:** Stock price prediction using past 24 hours
 - ▶ Input: Gold prices over 24 hours: [₹6,180, ₹6,200, . . . , ₹6,280]
 - ▶ Self-attention allows each hour to attend to all other hours
 - ▶ Example: Hour 24 might attend strongly to hours 17, 10, 3 (7-hour periodicity)
 - ▶ Captures temporal patterns automatically
- **Text Example:** "The student didn't pass the exam because **he** was unprepared"
 - ▶ Self-attention helps "he" attend to "student"
 - ▶ Resolves coreference within the sentence
 - ▶ Captures long-range dependencies ("he" $\rightarrow$ "student")

SELF-ATTENTION: VISUALIZATION

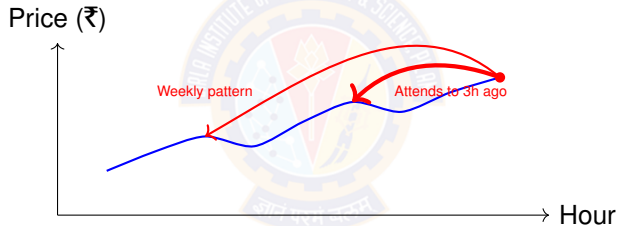
Text Example: Pronoun resolution with self-attention

Sentence: "The customer complained because he received wrong order"



SELF-ATTENTION: VISUALIZATION

Time Series: Each time step attends to relevant past time steps



CROSS-ATTENTION: INTER-SEQUENCE ALIGNMENT

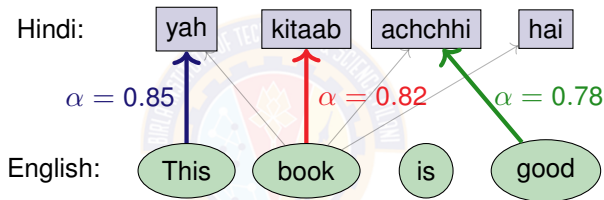
- **Cross-attention:** Query from one sequence, keys and values from *another*.

$$\mathbf{Q} = \mathbf{X}_A \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}_B \mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}_B \mathbf{W}_V$$

- **Time Series Example:** Multivariate forecasting
 - ▶ Sequence A: Stock prices (query current state)
 - ▶ Sequence B: Economic indicators (keys/values: inflation, GDP, interest rates)
 - ▶ Cross-attention: Stock prices attend to relevant economic indicators
 - ▶ Example: Gold price queries inflation rate and USD exchange rate
- **Text Example:** Hindi $\rightarrow$ English translation
 - ▶ Sequence A (Decoder): English words being generated
 - ▶ Sequence B (Encoder): Hindi source sentence
 - ▶ When generating "book", decoder attends to Hindi "kitaab"
 - ▶ Example: Generating "yah kitaab achchhi hai" $\rightarrow$ "This book is good"
 - ★ "book" (EN) attends to "kitaab" (HI) with $\alpha \approx 0.85$
 - ★ "good" (EN) attends to "achchhi" (HI) with $\alpha \approx 0.78$

CROSS-ATTENTION: VISUALIZATION

- Machine Translation Example: Hindi to English



- Key:** Decoder (English) dynamically attends to relevant parts of encoder (Hindi).
- This is exactly **Bahdanau attention** – the original cross-attention mechanism!

MASKED ATTENTION: CAUSAL/AUTOREGRESSIVE

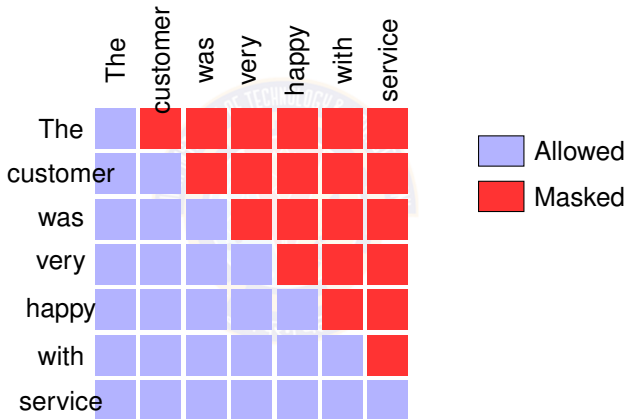
- **Masked attention:** Self-attention with *causal constraint* – can only attend to past.

$$\text{Mask}_{ij} = \begin{cases} 0 & \text{if } j \leq i \quad (\text{allowed}) \\ -\infty & \text{if } j > i \quad (\text{masked}) \end{cases}$$

- **Time Series Example:** Stock price prediction (prevent future leakage)
 - ▶ Predicting price at time t using only times $1, \dots, t-1$
 - ▶ Mask prevents attending to future times $t+1, t+2, \dots$
 - ▶ Ensures no "cheating" during training
 - ▶ Example: Predict tomorrow's gold price without seeing tomorrow's actual price
- **Text Example:** GPT-style language modeling
 - ▶ Task: Predict next word given previous words. "The customer was very ___"
 - ▶ Token at position 5 ("very") can only attend to positions 1-4
 - ▶ Cannot see future tokens during training or inference
 - ▶ Autoregressive: generate one token at a time, left-to-right

MASKED ATTENTION: CAUSAL MASK VISUALIZATION

- Attention mask pattern: Query (row) attends to Key (column)



- Interpretation: Lower triangular pattern ensures token i only sees tokens $\leq i$.
- Implementation: Add $-\infty$ to masked positions before softmax $\rightarrow$ weight = 0.

MULTI-HEAD ATTENTION: CONCEPT

- **Motivation:** Single attention may miss different types of relationships.
- **Solution:** Use multiple attention "heads" in parallel, each learning different patterns.
- **Time Series Example:** Gold price with 3 heads
 - ▶ **Head 1:** Attends to short-term trends (last 2-3 days)
 - ▶ **Head 2:** Attends to weekly patterns (7-day cycles)
 - ▶ **Head 3:** Attends to monthly patterns (30-day trends)
 - ▶ **Combined:** Richer representation than single attention
- **Text Example:** Sentence with 3 heads
 - ▶ **Syntax head:** Subject-verb-object relationships
 - ★ "The customer **complained**" – verb attends to subject
 - ▶ **Semantic head:** Meaning and coreference
 - ★ "he" attends to "customer" (resolves pronoun)
 - ▶ **Positional head:** Adjacent word dependencies
 - ★ "very happy" – modifier-adjective pair

MULTI-HEAD ATTENTION: ARCHITECTURE

- Formula (Vaswani et al., 2017):

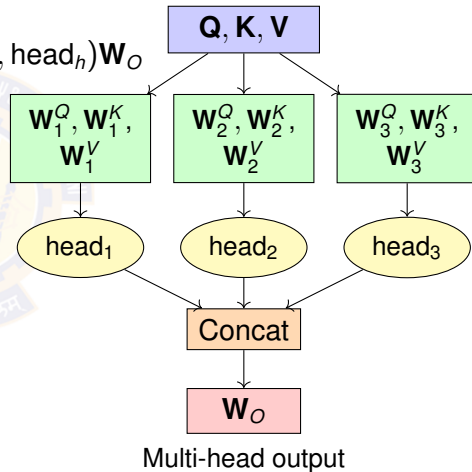
$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}_O$$

where each head i computes:

$$\text{head}_i = \text{Attention}(\mathbf{QW}_i^Q, \mathbf{KW}_i^K, \mathbf{VW}_i^V)$$

- Typical hyperparameters:

- ▶ $h = 8$ heads
- ▶ $d_{\text{model}} = 512$
- ▶ $d_k = d_v = 64$ per head



WHEN TO USE WHICH ATTENTION TYPE?

Attention Type	Use When...
Self-Attention	Modeling relationships within a single sequence Building contextual representations (e.g., BERT encoder) Capturing long-range dependencies in text or time series
Cross-Attention	Aligning two different sequences (seq2seq) Translation, summarization, question answering Transformer decoder attending to encoder outputs
Masked	Autoregressive generation (language modeling) Preventing information leakage from future GPT-style models, time series forecasting
Multi-Head	Capturing diverse relationship types simultaneously All modern Transformers (BERT, GPT, T5, ViT) When you need multiple attention patterns

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SELF-ATTENTION: FORMAL DEFINITION

- Self-attention allows each position to attend to all positions in the *same* sequence.
- Given input $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{n \times d}$:

- ① Linear projections:

$$\mathbf{Q} = \mathbf{XW}_Q, \quad \mathbf{K} = \mathbf{XW}_K, \quad \mathbf{V} = \mathbf{XW}_V$$

- ② Scaled dot-product attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}} \right) \mathbf{V}$$

- ③ Output:

$$\mathbf{Z} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \in \mathbb{R}^{n \times d_v}$$

- **Key property:** Every output $\mathbf{z}_i$ is a weighted combination of *all* inputs $\{\mathbf{x}_j\}_{j=1}^n$.
- **Difference from RNN:** Direct connections between all positions (no sequential bottleneck).

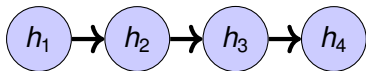
CNNs vs RNNs vs SELF-ATTENTION: COMPARIS

Aspect	CNN	RNN	Self-Attn
Receptive field	Local (kernel k)	Sequential (his- tory)	Global (all)
Path length	$O(\log n)$	$O(n)$	$O(1)$
Parallelization	High	Sequential	High
Complexity	$O(k \cdot n \cdot d^2)$	$O(n \cdot d^2)$	$O(n^2 \cdot d)$
Long-range	Many layers	Vanishing grad	Direct
Position info	Implicit	Built-in	Need encoding

- Key insight: Self-attention provides $O(1)$ path between any two positions!
- Trade-off: Better long-range modeling but $O(n^2)$ memory and computation.

PARALLELIZATION: RNN vs SELF-ATTENTION

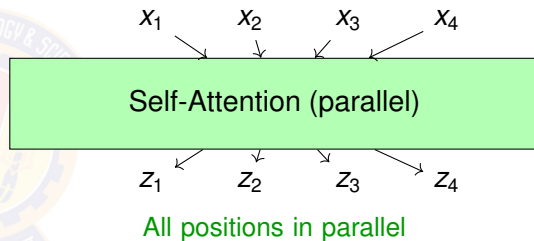
RNN (Sequential):



Must compute sequentially

Bottleneck: h_4 depends on h_3 depends on h_2 depends on h_1

Self-Attention (Parallel):



Benefit: All outputs computed simultaneously on GPU!

Result: Transformers train 10-100× faster than RNNs for long sequences.

POSITION-INVARIANCE PROBLEM

- Critical issue: Self-attention is *permutation-invariant*!

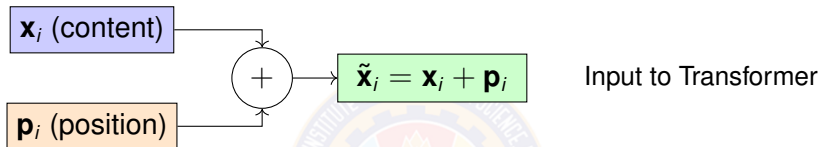
$$\text{Attention}(\mathbf{X}) = \text{Attention}(\mathbf{PX})$$

for any permutation matrix $\mathbf{P}$.

- Example: These sentences are treated identically:
 - ▶ Original: "The customer was happy with service"
 - ▶ Shuffled: "service with happy was customer The"
- Why? Attention weights depend only on *content*, not *position*.
- Problem for NLP: Word order matters!
 - ▶ "Dog bites man" $\neq$ "Man bites dog"
 - ▶ Need position information to distinguish
- Solution: Inject positional information into input embeddings.

POSITIONAL ENCODING: THE SOLUTION

Add position information to input embeddings:



Two approaches:

- ① **Learned positional embeddings** (BERT, GPT-2):
 - ▶ $\mathbf{p}_i$ is a learned parameter for each position i
 - ▶ Simple, but doesn't generalize beyond training length
- ② **Fixed sinusoidal encoding** (Original Transformer):
 - ▶ $\mathbf{p}_i$ computed using sine/cosine functions
 - ▶ Deterministic, generalizes to any sequence length

ABSOLUTE POSITIONAL ENCODING: SINUSOIDAL

Vaswani et al. (2017): Use fixed sinusoidal functions.

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$$

$$\text{PE}(\text{pos}, 2i + 1) = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$$

- pos: Position in sequence (0, 1, 2, ...)
- i : Dimension index (0, 1, ..., $d_{\text{model}}/2 - 1$)
- d_{model} : Model dimension (e.g., 512)

Intuition:

- Different dimensions use different frequencies
- Lower dimensions = lower frequency (long-range patterns)
- Higher dimensions = higher frequency (fine-grained positions)
- Each position gets unique encoding vector

WHY SINUSOIDAL ENCODING WORKS

- **Key mathematical property:** For any fixed offset k , $\text{PE}(\text{pos} + k)$ is a *linear function* of $\text{PE}(\text{pos})$.

$$\text{PE}(\text{pos} + k, 2i) = \mathbf{M}_k \cdot \text{PE}(\text{pos}, 2i) \quad \text{for some transformation matrix } \mathbf{M}_k$$

- **Trigonometric identity:**

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

- **Example:** For dimension i , let $\omega_i = 1/10000^{2i/d}$

$$\text{PE}(\text{pos} + k, 2i) = \sin((\text{pos} + k)\omega_i) = \sin(\text{pos}\omega_i) \cos(k\omega_i) + \cos(\text{pos}\omega_i) \sin(k\omega_i)$$

This is a linear combination of $\text{PE}(\text{pos}, 2i)$ and $\text{PE}(\text{pos}, 2i + 1)$.

- **Benefit:** Model can learn to attend by *relative position* (e.g., "3 positions back").

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SUMMARY: ATTENTION MECHANISMS

① Attention Mechanism

- ▶ Dynamic, weighted aggregation based on relevance
- ▶ Query-Key-Value framework for information retrieval
- ▶ Solves bottleneck by attending selectively

② Scoring Functions

- ▶ Dot-product: simple, fast
- ▶ Scaled dot-product: prevents saturation (Transformer default)
- ▶ Additive: more expressive, slower

③ Attention Types

- ▶ Self: intra-sequence, Cross: inter-sequence
- ▶ Masked: causal, Multi-head: diverse patterns

④ Positional Encoding

- ▶ Required for position-aware self-attention
- ▶ Sinusoidal: fixed, generalizable

KEY FORMULAS TO REMEMBER

- 1. Attention (general):

$$\text{Attention}(\mathbf{q}, \{\mathbf{k}_i, \mathbf{v}_i\}) = \sum_{i=1}^n \alpha_i \mathbf{v}_i \quad \text{where} \quad \alpha_i = \text{softmax}(\text{score}(\mathbf{q}, \mathbf{k}_i))$$

- 2. Scaled Dot-Product Attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}$$

- 3. Multi-Head Attention:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}_O$$

- 4. Positional Encoding:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right), \quad \text{PE}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right)$$

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Thank You!